The University of Alabama at Birmingham

Introduction to Machine Learning

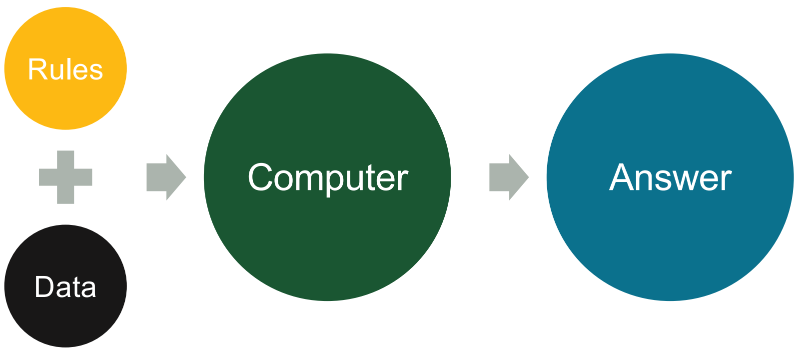
Keren Li, PhD
Department of Mathematics
University of Alabama at Birmingham
@ Data Science UAB Summer Camp

Introduction to Machine Learning

Can computers learn from experience?

What is Machine Learning?

Traditional Programming



Rules:

- If score ≥ 90 → A
- If score ≥ 80 → B

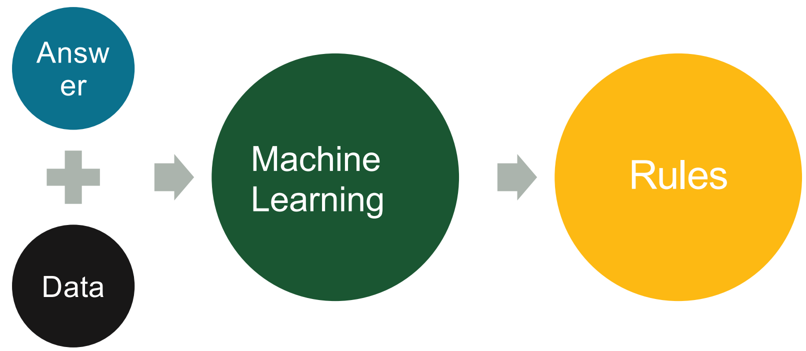
Data:

- Student score = 85

Answer:

- B

Machine Learning



Data:

- Many student records

Answers:

- Final grades

Machine learns:

- Patterns relating scores to grades

Machine Learning learns patterns from examples.

Why Machine Learning?

Can we write rules for everything?

Face Recognition

🙂 → 👤

Unlocking your phone

Spam Detection

📧 → 🚫

Filtering unwanted email

Self-Driving Cars

🚗

Understanding roads and traffic

Some problems are too complex for humans to program manually.

Example: Cat vs Dog

How would you teach a computer to tell cats and dogs apart?

What rules would you write?

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Too many exceptions!

Let the computer learn from examples instead.

Learning from Examples

How does a computer learn?

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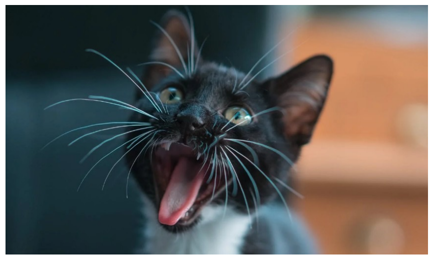

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→ Cat

→ Dog

Many Examples

↓

Machine Learning

↓

Learned Model → Prediction

New Image

↓

Prediction

↓

Cat/Dog ?

Machine learning learns patterns from labeled examples.

Types of Machine Learning

Supervised Learning

Learn from examples with known answers

- House prices
- Disease prediction
- Spam detection

Unsupervised Learning

Find patterns without answers

- Customer groups
- Music recommendations
- Community detection

Reinforcement Learning

Learn through rewards and mistakes

- Chess
- Robotics
- Self-driving systems

Different problems require different ways of learning.

Supervised Learning

Learning from examples with known answers

Study Hours

Attendance

Homework

↓

Machine Learning

↓

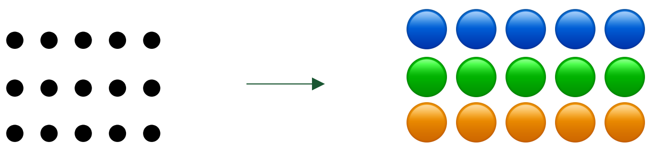
Exam Score

Examples:

- House Price Prediction
- Disease Prediction
- Spam Detection
- College Admissions

Unsupervised Learning

Finding patterns without known answers



No answers are provided.
The machine discovers patterns itself.

Real-World Examples

- **Customer Segmentation** 🛒
  Different shopping habits
- **Music Recommendations** 🎵
  Similar songs grouped together
- **Social Networks** 👥
  Communities and friend groups

Reinforcement Learning

Learning through trial and error

Action

↓

World

↓

Reward / Penalty

↓

Learning

↓

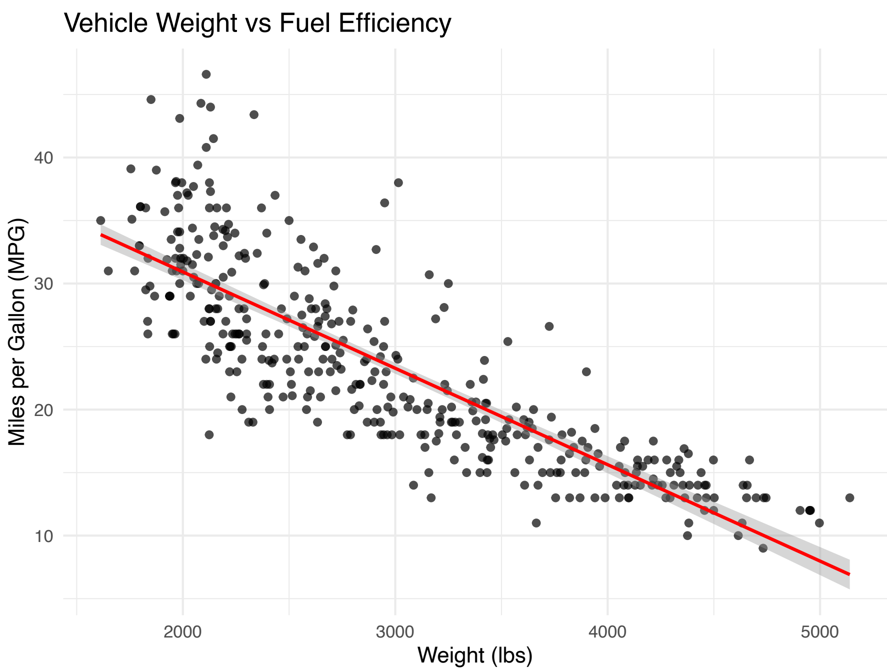
Better Action

Examples

- **Chess** ♟️
  Win = Reward
  Lose = Penalty
- **Robotics** 🤖
  Walk successfully = Reward
  Fall down = Penalty
- **Game AI** 🎮
  Higher score = Reward
  Lower score = Penalty

The machine learns by trying, making mistakes, and improving over time.

A Simple Prediction Model



Input:	Example:
Vehicle Weight	Weight = 3500 lbs
↓	
Machine Learning Model	
↓	
Prediction:	Predicted MPG ≈ 20
Fuel Efficiency (MPG)	

Machine learning learns patterns from historical data and uses them to predict future outcomes.

Training and Testing

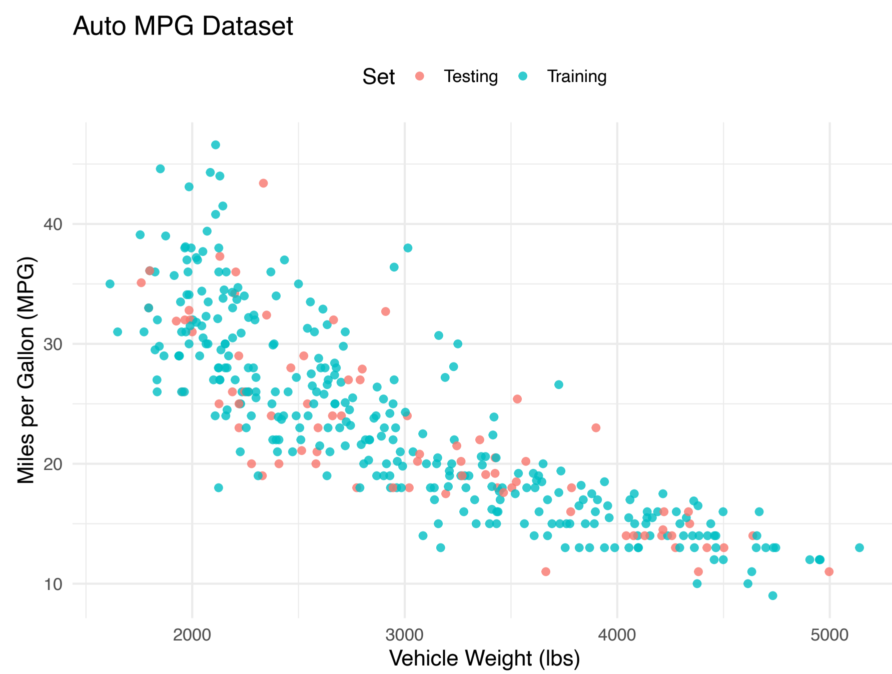
Auto MPG Dataset
(392 Cars)

Training Data
80%
314 Cars

Learn Pattern

Testing Data
20%
78 Cars

Evaluate Accuracy



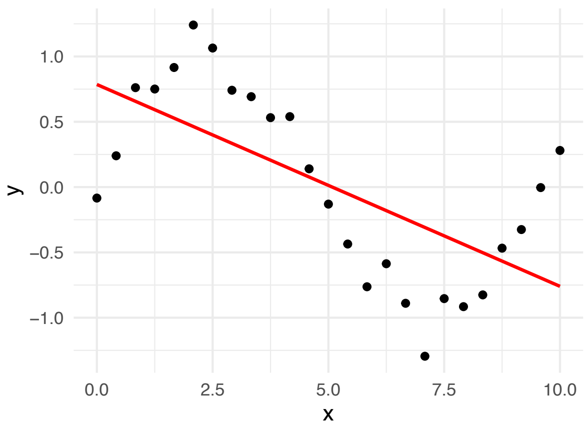
Can the model predict MPG for cars it has never seen before?

Overfitting

When a model memorizes instead of learning

Underfitting

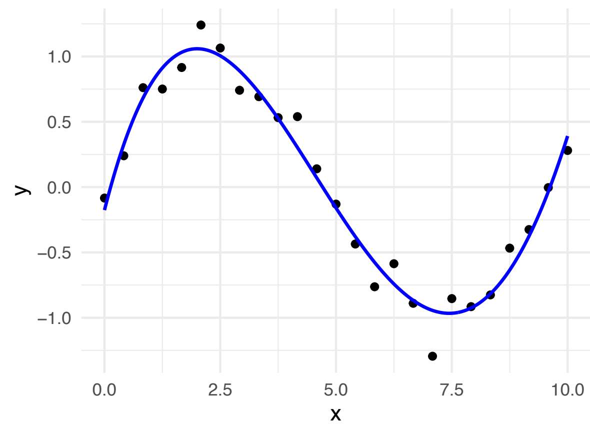
Too Simple



Misses important patterns

Good Fit

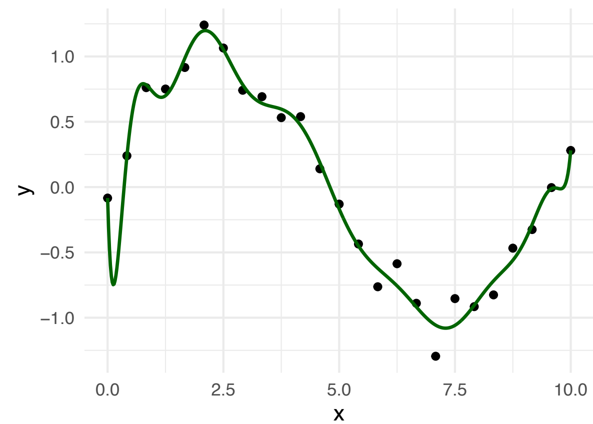
Just Right



Captures the main relationship

Overfitting

Too Complex



Memorizes noise

Good models learn patterns, not individual observations.

What Makes a Good Model?

Accuracy 🎯

Makes correct predictions

Interpretability 🔍

Can humans understand it?

Speed ⚡

Can it make predictions quickly?

Fairness ⚖️

Treats people fairly

The most accurate model is not always the best model.

Machine Learning Around You

You already use machine learning every day.

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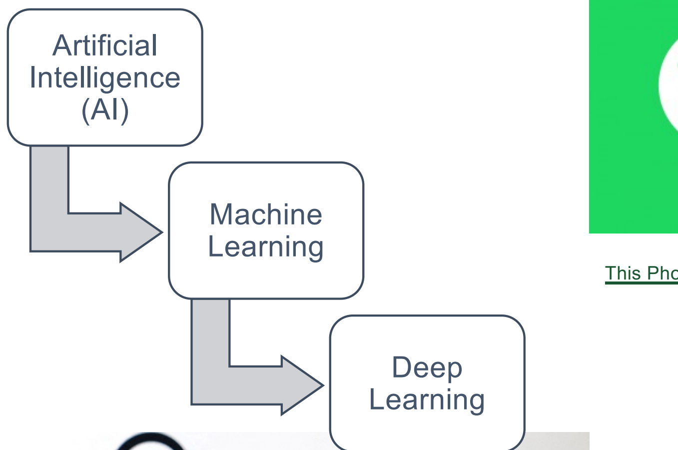

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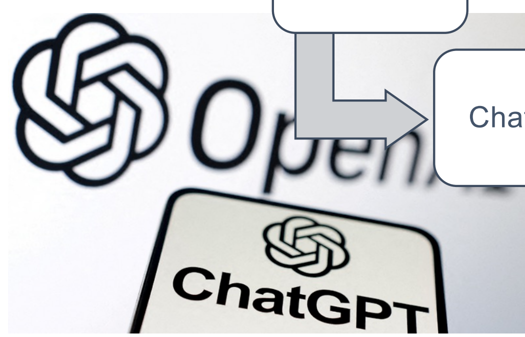

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The Future of AI and Data Science

Opportunities
- 🏥 Medicine
- 🎓 Education
- 🔬 Science
- 🌎 Climate Research

Future Careers
- Data Scientist
- Statistician
- Machine Learning Engineer
- Biostatistician
- Actuary
- Quantitative Analyst

Challenges
- ⚖️ Fairness
- 🔒 Privacy
- 📰 Misinformation

Key Takeaways

Machine Learning:

- Learns patterns from data
- Makes predictions
- Improves through experience
- Powers many technologies we use every day
- Works best when combined with statistics, computing, and domain knowledge

Machine learning is not replacing human thinking.

It is a tool that helps people make better decisions using data.